

*MA.7.AR.3.1***Benchmark****MA.7.AR.3.1 Apply previous understanding of percentages and ratios to solve multi-step real-world percent problems.**

Example: 23% of the junior population are taking an art class this year. What is the ratio of juniors taking an art class to juniors not taking an art class?

Example: The ratio of boys to girls in a class is 3: 2. What percentage of the students are boys in the class?

Benchmark Clarifications:

Clarification 1: Instruction includes discounts, markups, simple interest, tax, tips, fees, percent increase, percent decrease and percent error.

Connecting Benchmarks/Horizontal Alignment

- MA.6.AR.3.2, MA.6.AR.3.4
- MA.7.NSO.2.1, MA.7.NSO.2.2, MA.7.NSO.2.3
- MA.7.DP.1.3
- MA.7.DP.2.2, MA.7.DP.2.3, MA.7.DP.2.4

Terms from the K-12 Glossary

- Percent of Change
- Percent Error
- Rate
- Simple Interest

Vertical Alignment**Previous Benchmarks**

- MA.5.FR.1.1
- MA.5.AR.1.2, MA.5.AR.1.3

Next Benchmarks

- MA.7.AR.4.4, MA.7.AR.4.5
- MA.8.AR.2.1

Purpose and Instructional Strategies

In previous courses, students represented the division of two whole numbers as a fraction. In doing this, students started to work with a ratio relationship that relates parts to wholes. In grade 6 accelerated, students solved mathematical and real-world problems involving percentages, ratios, rates and unit rates. Students also solve multi-step real-world percent problems. In future courses, students solve multi-step linear equations given any context.

- Instruction includes discounts, markups, simple interest, tax, tips, fees, percent increase, percent decrease and percent error (*MTR.7.1*).
 - Markdown/discount is a percentage taken off the initial price. Instruction includes showing the connection between subtracting the calculated discount or taking the difference between 100% and the discount and multiplying that by the initial price.
 - For example, if there was a 15% discount on an item that costs \$15.99, students could take 85% of \$15.99 or take 15% of \$15.99 and subtract that value from the initial price of \$15.99.

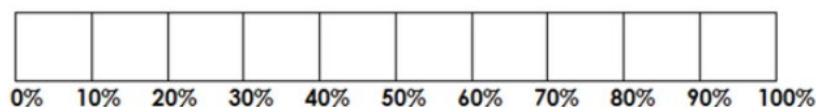
- Markup showcases adding a charge to the initial price. Markups are often shown in retail situations.
- Simple interest refers to money you can earn by initially investing some money (the principal). The percentage of the principal (interest) is added to the principal, making your initial investment grow. The formula, $I = Prt$, represents I =interest, P =principal, r =rate, and t =time. When using simple interest, provide the formula as students should not be expected to memorize this.
- Tax, tips and fees are an additional charge added to the initial price. Students can add the calculated tax, tip or fee to the initial price or add 1 to the tax, tip or fee to reach the final cost.
 - For example, if there was a 6% sales tax on clothing and a t-shirt costs \$7.99. Students can add 100% to the 6% and multiply that value to \$7.99 or students can find 6% of the \$7.99 and add that to the initial value of the t-shirt.
- Percent Increase/Percent Decrease asks students to look for a percentage instead of a dollar amount. Students should discover that they can use the formula below to help become more flexible in their thinking.

$$\frac{\text{new price} - \text{initial price}}{\text{original price}} \times 100$$

- Percent Error is a way to express the size of the error (or deviation) between two measurements.

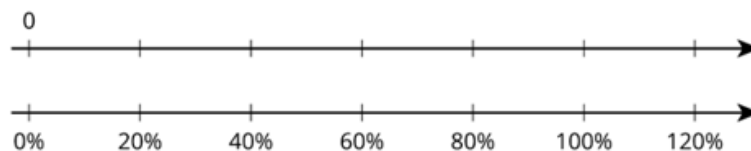
$$\% \text{ error} = \frac{|\text{estimation} - \text{actual}|}{\text{actual}} \times 100$$

- Use bar models to model percent increase and decrease problems.
 - For example, if you are finding percentages that are in multiples of 10%, your bar model may look like the model below.



To showcase the percent increase, you would add additional boxes into the bar model. If you are showcasing a percent decrease, then you would cross out boxes for the decrease (*MTR.2.1*).

- Use bar models, double number lines, tables or other visual representations to model relationships between percentages and the part and whole amounts (*MTR.2.1*).
 - Double Number Line



- Table

Example: Calculate 10%, 20%, 5%, 15%, 25%, & 1% of 300

Percent	100	10	20		
Total	300	30	60		

Instruction includes the use of patterns when using a table. In the example above, students can use the idea of 100% being 300 and using this knowledge to find other percentages. 10 is $\frac{1}{10}$ of 100, so students can divide by 10. To find 20%, students can multiply their solution from 10% by 2. The pattern can continue to relate common connections between percentages (*MTR.5.1*).

- Reinforce how percentages relate to fractions and decimals. Help students write equivalent ratios to represent problems using reasoning about the relationships between the quantities.
 - Instruction includes using proportional relationships and multiplicative reasoning to solve problems.

Common Misconceptions or Errors

- Students may incorrectly place the decimal point when calculating with percentages. If students have discovered the shortcut of moving the decimal point twice, instruction includes understanding of how a percent relates to fractions and decimals. Refer to MA.7.NSO.1.2 to emphasize equivalent forms.
- Students may forget to change the percent amount into decimal form (divide the percent by 100) when setting up an equation (*MTR.3.1*).
- Students may incorrectly believe all percentages must be between 1 and 100%. To address this misconception, provide examples of percentages below 1% and over 100%.
- Students may incorrectly believe a percent containing a decimal is already in decimal form.
 - For example, emphasize that 43.5% is 43.5 out of 100 and dividing by 100 will provide the decimal form.
- In multiple discount problems, students may incorrectly combine the discounts instead of working them sequentially (*MTR.5.1*).
 - For example, 25% off, then 10% off could incorrectly lead to 35% off rather than finding 25% off before calculating the additional 10% off.
- Students may incorrectly invert the part and the whole in the percent problem. To address this misconception, students should use bar models to help visualize and make sense of the problem (*MTR.2.1*).

Strategies to Support Tiered Instruction

- Instruction includes the use of estimation to find the approximate solution before calculating the actual result to help with correct placement of the decimal point and reasonableness of the solution.
- Teacher provides opportunities for students to use a 100 frame to review place value for and the connections to decimal, fractional, and percentage forms.

- Teacher provides support for students in dividing by 100 to change percent into decimal form. Teacher supports by providing calculators, manipulatives and base ten blocks to multiply decimals.
- Instruction includes having students take different percentages of the same amount, such as 40% of 80, 4% of 80, 0.4% of 80, 0.04% of 80 and 400% of 80. Students can be given the flexibility to provide the answer as decimal or fraction and compare.
- Teacher provides support for students when solving multi-discount problems and combining the discounts. Instruction might begin with a single-step discount problem in a real-world context.
 - For example, the teacher can include local sale flyers with products that students are interested in buying. Have students explain how to apply the multi-discounts with a comparison of the difference in costs when combining the discounts incorrectly.
- Teacher provides opportunities for students to reason and think about multiple discount problems by providing prompts.
 - For example, “if a pair of jeans is 50% off with an additional 50% off, does that mean the jeans are 100% off, or free?” or “what if the jeans are 75% off with an additional 50% off, does that mean the jeans are 125% off and the store now owes you money to take them?”
- Teacher provides opportunities for students to comprehend the context or situation by engaging in questions (laminating these questions on a printed card for students to utilize as a resource in and out of the classroom would be helpful).
 - What do you know from the problem?
 - What is the problem asking you to find?
 - Can you create a visual model to help you understand or see patterns in your problem?
- Teacher provides support when solving multi-discount problems, by providing students with a table to keep track of the information in the problem.
- Instruction includes the use of a three-read strategy. Students read the problem three different times, each with a different purpose (laminating these questions on a printed card for students to utilize as a resource in and out of the classroom would be helpful).
 - First, read the problem with the purpose of answering the question: What is the problem, context, or story about?
 - Second, read the problem with the purpose of answering the question: What are we trying to find out?
 - Third, read the problem with the purpose of answering the question: What information is important in the problem?
- Teacher encourages the use of bar models to help visualize and make sense of the problem.
- Instruction includes understanding of how a percent relates to fractions and decimals if students have discovered the shortcut of moving the decimal point twice. Refer to MA.7.NSO.1.2 to emphasize equivalent forms.

Instructional Tasks

Instructional Task 1 (MTR.4.1, MTR.7.1)

SurfPro Shop and The Surfer Store both sold surfboards for \$350. In February, SurfPro Shop wanted to increase their profits, so they increased the prices of their boards by 15%. When this increase failed to bring in more money, they decreased their price again by 10% in November. To beat their competitor who had increased prices, The Surfer Store decided to decrease their price of surfboards by 10% in March. However, when they started to lose money on the final pricing scheme, they increased the price of surfboards in November by 15%.

Part A. If no other changes were made after November, which store now has the better price for surfboards?

Part B. What is the difference between their prices?

Instructional Items

Instructional Item 1

A college's intramural soccer team has 30 players, 60% of which are women. After 22 new players joined the team, the percentage of women was reduced to 50%. How many of the new players are women?

Instructional Item 2

Miguel takes out a loan that adds interest each year on the initial amount. What is the interest Miguel will pay on the loan if he borrowed \$5,000 at an annual interest rate of 4.5% for 15 years? (Use the formula $I = Prt$, where I is the interest, P is the principal or initial investment, r is the interest rate per year, and t is the number of years.)

Instructional Item 3

Massimo lost his mathematics textbook. The school charges a lost book fee of 70% of the initial cost of the book. If Massimo received a notice that he owed the school \$73.50 for the lost textbook, what was the initial cost?

**The strategies, tasks and items included in the BIG-M are examples and should not be considered comprehensive.*